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# Population Pharmacokinetics of Empagliflozin, a Sodium Glucose Cotransporter 2 Inhibitor, in Patients With Type 2 Diabetes

**Matthew M. Riggs, PhD[1a], Alexander Staab, PhD[2], Leo Seman, MD[3], Thomas R. MacGregor, PhD[3], Timothy T. Bergsma, PhD[1], Marc R. Gastonguay, PhD[1a], and Sreeraj Macha, PhD[3]**

## Abstract

Data from five randomized, placebo-controlled, multiple oral dose studies of empagliflozin in patients with type 2 diabetes mellitus (T2DM; N = 974; 1–100 mg q.d.; ≤12 weeks) were used to develop a population pharmacokinetic (PK) model for empagliflozin. The model consisted of two-compartmental disposition, lagged first-order absorption and first-order elimination, and incorporated appropriate covariates. Population estimates (interindividual variance, CV%) of oral apparent clearance, central and peripheral volumes of distribution, and inter-compartmental clearance were 9.87 L/h (26.9%), 3.02 L, 60.4 L (30.8%), and 5.16 L/h, respectively. An imposed allometric weight effect was the most influential PK covariate effect, with a maximum effect on exposure of ±30%, using 2.5th and 97.5th percentiles of observed weights, relative to the median observed weight. Sex and race did not lend additional description to PK variability beyond allometric weight effects, other than ~25% greater oral absorption rate constant for Asian patients. Age, total protein, and smoking/alcohol history did not affect PK parameters. Predictive check plots were consistent with observed data, implying an adequate description of empagliflozin PKs following multiple dosing in patients with T2DM. The lack of marked covariate effects, including weight, suggests that no exposure-based dose adjustments were required within the study population and dose range.

## Keywords

empagliflozin, pharmacokinetics, diabetes, SGLT2

Type 2 diabetes mellitus (T2DM) is a chronic disease that affects an estimated 285 million people worldwide, and is characterized by hyperglycemia as a result of insulin resistance and impaired insulin secretion.[1] Hyperglycemia can cause complications in many organ systems, including the nervous and vascular systems.[2–4]

Approaches to lower blood glucose in patients with T2DM include diet and exercise modifications, with the addition of oral anti-hyperglycemic medications and insulin, if needed. In the UK Prospective Diabetes Study (UKPDS), intensive treatment with metformin, sulfonylureas, or insulin resulted in only a limited improvement in glycemic control over 10 years compared with dietary modification alone, with a significant loss in glycemic control over time.[5] Use of available therapeutic agents may also be limited by clinically important side effects such as hypoglycemia, weight gain, and edema.[6,7] These limitations indicate a need for additional anti-hyperglycemic therapeutic options for patients with T2DM.[8]

The sodium glucose cotransporter (SGLT)2 is located in the brush border membrane of the proximal convoluted tubule of the nephron and is responsible for the reabsorption of glucose from the glomerular filtrate.[8] Under normoglycemic conditions, glucose is completely reabsorbed by SGLTs in the kidney. However, the reuptake capacity of the kidney is saturated at glucose concentrations of approximately 11 mmol/L, resulting in glucosuria. This threshold concentration can be decreased by inhibition of SGLT2.[8,9] Several SGLT2 inhibitors are in development as potential treatments for T2DM. Due to their insulin-independent mode of action, SGLT2 inhibitors are expected to be associated with a low risk of hypoglycemia and have the potential to be combined with other anti-hyperglycemic drugs.[8]

Empagliflozin is a potent and highly selective oral SGLT2 inhibitor that has been shown to inhibit renal glucose reabsorption in patients with T2DM in Phase I studies, leading to increased urinary glucose excretion and decreased plasma glucose levels,[10,11] consistent with findings with other SGLT2 inhibitors.[12,13]

[1]Metrum Research Group LLC, Tariffville, CT, USA
[2]Boehringer Ingelheim Pharma GmbH & Co. KG, Biberach, Germany
[3]Boehringer Ingelheim Pharmaceuticals, Inc., Ridgefield, CT, USA

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**Corresponding Author:**
Matthew M. Riggs, PhD, Metrum Research Group LLC, 2 Tunxis Road, Suite 112, Tariffville, CT 06081, USA
Email: mattr@metrumrg.com
[a]Members of the American College of Clinical Pharmacology.

The pharmacokinetics (PKs) of empagliflozin in patients with T2DM were similar following single and multiple doses at steady-state using noncompartmental analyses. Peak levels were reached 1.5–3.0 hours after oral administration.[10] Plasma concentration-time profiles showed a biphasic decline, that is, a rapid disposition phase and a slower disposition phase. The mean terminal elimination half-life was similar after single dose and at steady-state, ranging from 10 to 19 hours.[10] Increases in empagliflozin exposure were approximately dose proportional from 2.5 to 100 mg q.d. empagliflozin.[10] The amount of parent drug excreted unchanged in the urine ranged from 12% to 19% of the administered dose at steady-state.[10] Similarities in single-dose and steady-state parameters suggest linear PKs with respect to time.[10] Up to 23% drug accumulation was observed at steady-state.[10]

The aim of this analysis was to develop a population PK model from Phase I and II data to estimate the effects of covariates, such as demographics, patient habits, and laboratory values, which may explain variability in empagliflozin PK parameters.

## Methods

### Patients and Sampling

Data were obtained from five randomized Phase I and II clinical trials (*EudraCT No. 2007-000654-32*,[10] *NCT00558571*,[11] *NCT00885118*, *NCT00789035*,[14] and *NCT00749190*; referred herein as *Studies A*, *B*, *C*, *D*, and *E*, respectively). All clinical trial protocols were approved by the relevant local Institutional Review Board or Independent Ethics Committee and were carried out in compliance with the Ethical Principles for Medical Research Involving Human Subjects of the Declaration of Helsinki (October 1996), the International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use tripartite harmonized guideline E6 (R1) for Good Clinical Practice, and the standard operating procedures of Boehringer Ingelheim. In Study A, patients with T2DM were randomized to receive placebo or empagliflozin in doses of 2.5, 10, 25, or 100 mg q.d. over 8 days. In Study B, patients with T2DM were randomized to receive placebo or empagliflozin in doses of 10, 25, or 100 mg q.d. over 4 weeks. In Study C, Japanese patients with T2DM were randomized to receive either placebo or empagliflozin 1, 5, 10, or 25 mg q.d. over 4 weeks. In Study D, patients with T2DM were randomized to receive empagliflozin 5, 10, 25 mg q.d., placebo, or open-label metformin 1,000 mg b.i.d. (or maximum tolerated dose) over 12 weeks. In Study E, patients with T2DM were randomized to empagliflozin 1, 5, 10, 25, or 50 mg q.d., placebo, or open-label sitagliptin 100 mg q.d., added to metformin for 12 weeks. All the included studies were double-blind and placebo-controlled. Blood samples for the determination of serum empagliflozin concentrations were collected at selected times.

### Determination of Plasma Empagliflozin Concentrations

For quantification of empagliflozin plasma concentrations, 3 mL of blood was taken from a forearm vein in an ethylendiaminetetraacetic acid (EDTA)-anticoagulant blood drawing tube at the specified times. The EDTA-anticoagulated blood samples were centrifuged immediately after collection (or within approximately 30 minutes after being stored on ice) for approximately 10 minutes (at approximately 2,000–4,000*g*) at 4–8°C. Plasma samples were stored at $\leq -20$°C. Empagliflozin concentrations in K3-EDTA plasma samples were quantified using a validated high-performance liquid chromatography-tandem mass spectrometry (HPLC-MS/MS) assay. Empagliflozin and the internal standard $[^{13}C_6]$-empagliflozin were extracted from plasma by supported liquid extraction (SLE). After evaporation under nitrogen, the residue was reconstituted and analyzed using validated HPLC-MS/MS assays. The lower limit of quantification for empagliflozin in plasma was 1.11 nmol/L, with linearity to 1,110 nmol/L. The mean inaccuracy and imprecision (%CV) of calibration standards were $<\pm 1.6\%$ relative error and $<8.1\%$ relative standard deviation, respectively. Results were calculated using peak area ratios and calibration curves were created using weighted ($1/x^2$) quadratic regression.

### Data Analysis

The PK data were analyzed using nonlinear mixed-effects modeling with the NONMEM® software system Version VI Level 2.0. The pharmacostatistical model structure was guided by earlier population PK modeling of combined Phase I and II data (unpublished).

This previous population PK modeling for empagliflozin indicated that a two-compartment dispositional model, with lagged first-order oral absorption and first-order elimination from the central compartment, could appropriately describe empagliflozin plasma concentration data. In addition, the empagliflozin oral absorption profile from early studies was similar to the reported profiles for glibenclamide, moxonidine, furosemide, and amiloride, for which transit compartment models had been used to describe their absorption profile.[15] Therefore, exploratory modeling investigated an alternative absorption model based on transit compartments to describe the empagliflozin plasma concentration-time profile at early time post-dose observations. Three transit compartments, with an estimated mean transit time of approximately 30 minutes, were included in series to precede a first-order absorption compartment. This model provided acceptable fits to the PK data, including the first several hours post-dose.

The computation time, however, for the full covariate model without the transit compartments (absorption lag time [ALAG] fixed at 0.5 hours) and without the PK observations from the first hour post-dose (including a $COVARIANCE step) was nearly 40 times faster than the transit compartment model without $COVARIANCE (approximately 45 minutes compared to >29 hours). The parameter estimates using the simplified model (ALAG fixed at 0.5 hours) and data set were comparable to the previous final model, and interpretations of the covariate effects were unchanged. No covariates associated with ALAG were investigated. Therefore, the improvement in computation time was achieved with no notable loss of precision or increase in bias for the covariate estimates, indicating that this simplified model may be appropriately applied for subsequent empagliflozin population PK modeling. In addition, this comparison showed that collections from early observation times (which are unlikely to be available during subsequent clinical trials) were not necessary for proper estimation of the included covariate effects on empagliflozin oral clearance, volume of distribution and the absorption rate constant. Consequently, the absorption lag in the current analysis was fixed at 0.5 hours and only observations collected after 1 hour post-dose were included.

This model was implemented using the PREDPP subroutine ADVAN4 TRANS4, which accounts for multiple dosing or steady-state conditions using recursive superposition. Differences between the observed ($C_{ij}$) and predicted ($\hat{C}_{ij}$) empagliflozin plasma concentrations were quantified using interindividual random-effect distributions, modeled using exponential variance models with covariance terms between the apparent oral PK parameters (CL/F, V3/F, KA) and residual random effects, described with a combined additive ($\varepsilon_{a,ij}$) and proportional ($\varepsilon_{p,ij}$) model. Separate residual variance terms were included during model development to account for the greater variability in the empagliflozin concentrations observed from *Studies D* and *E* compared with *Studies A*, *B*, and *C*.

The first-order conditional estimation method with $\eta$–$\varepsilon$ interaction (FOCE-INT) was employed for all model runs. Objective function values and goodness-of-fit plots were used as model selection criteria. Diagnostic plots and post-processing of NONMEM outputs were performed using R (version 2.10.1).[16]

An attempt was made to incorporate known physiologic relationships into the covariate–parameter models. For example, an allometric model (equation 1) describes the change in physiologic parameters as a function of body size both theoretically and empirically.[17–21] In cases where no physiologic relationship was known a priori, the effects of continuous covariates were modeled using a normalized power model while the effect of categorical covariates was similarly described (equation 1).

$$TVP = \theta_n \cdot \left(\frac{WT_i}{WT_{ref}}\right)^{\theta_{allo}} \cdot \prod_{1}^{m} \left(\frac{cov_{mi}}{ref_m}\right)^{\theta_{(m+n)}} \cdot \prod_{1}^{p} \theta_{(p+m+n)}^{cov_{pi}} \qquad (1)$$

The typical value of a model parameter (TVP) was described as a function of individual body weight ($WT_i$) normalized by a reference weight ($WT_{ref}$; which is the mean for the population), m individual continuous covariates ($cov_{mi}$), and p individual categorical (0–1) covariates ($cov_{pi}$) such that $\theta_n$ is an estimated parameter describing the typical PK parameter value for an individual with covariates equal to the reference covariate values ($WT_i = WT_{ref}$, $cov_{mi} = ref_m$, $cov_{pi} = 0$). $\theta_{(m+n)}$ and $\theta_{(p+m+n)}$ are estimated parameters describing the magnitude of the covariate–parameter relationships. The fixed effect parameter $\theta_{allo}$ is a fixed allometric power parameter, which was assigned a value of 0.75 for physiologic processes, such as clearances, and a value of 1 for anatomical volumes (equation 1).

A covariate modeling approach emphasizing parameter estimation was implemented to assess the clinical relevance of covariate effects and explain the apparent absence of covariate effects (true lack of an effect versus lack of information about that effect). Predefined covariate–parameter relationships were identified based on exploratory graphics, scientific interest, and mechanistic plausibility based on prior knowledge. A full model was constructed with care to avoid correlation or colinearity in predictors.

Nearly half of the subjects from *Studies D* and *E* were female, allowing for the inclusion of sex in the covariate analysis. Estimated creatinine clearance was correlated with a subject's weight and age. Therefore, a simple inverse relationship with serum creatinine was used as a surrogate for renal filtration-associated clearance. Laboratory tests were only included in the full covariate analysis if there was a plausible mechanism for their influence on PK variability. For example, total protein may influence the fraction of drug bound to plasma protein and so was considered as a potential covariate. In total, the full covariate model included the allometric weight terms, patient age, sex, race and total protein on CL/F, V2/F, and V3/F. Race was specifically investigated for exposure differences between the Asian (Japanese, Taiwanese, and Koreans) and non-Asian patients. Additional covariates evaluated on CL/F were smoking and alcohol histories and serum creatinine. An allometric weight term was included on Q/F and the oral absorption rate constant included an effect for Asian race. Of note, sex and Asian race were expected to have potential correlations with body weight, for example, Asian patients would weigh less than non-Asian patients, on average. Fixing the weight covariate effects using the "allometric" scaling was performed to overcome such potential correlation and allowed for the investigation of any additional influence, beyond that described by weight, of sex and Asian race on the empagliflozin PK parameters.

To correct for potential variation in the normal reference range of laboratory values (due to laboratorydependent factors) when pooling data, laboratory values were converted to a standard unit of measurement and normalized across a standard reference range. As a result, some laboratory value ranges were not necessarily bounded to minimal values of zero. Normalization was done via a linear transformation of the standard reference range according to equation (2).

$$\langle \text{normalized} \rangle = \mathrm{L} + (\langle \text{converted} \rangle - \mathrm{l}) * \frac{\mathrm{H} - \mathrm{L}}{\mathrm{h} - \mathrm{l}} \tag{2}$$

where H and L denote the higher and lower boundaries of the standard reference range, respectively; and h and l denote the higher and lower boundaries of converted reference range, respectively.

Population parameters, including fixed-effects parameters (covariate coefficients and structural model parameters) and random-effects parameters, were estimated. Inferences about clinical relevance of parameters were based on the resulting parameter estimates of the full model and measures of estimation precision (asymptotic standard errors and bootstrap 95% confidence intervals [CIs]).

The adequacies of the final model and parameter estimates were investigated with a predictive check method assuming that parameter uncertainty is negligible relative to interindividual and residual variance.[22,23] The minimum, median, and maximum observed concentration values were identified for each subject in the observed dataset (N = 974). The deciles (10th, 20th, 30th, 40th, 50th [median], 60th, 70th, 80th, 90th percentiles) of the 974 observed values were then determined for each statistic (minimum, median, and maximum). This was repeated for each of the 300 Monte Carlo simulated datasets, resulting in 300 simulated values of each percentile (for each statistic). For each statistic, distributions of the 300 simulations of each percentile were compared graphically with the unique value of each percentile from the observed dataset. Within the plots and for each percentile, the simulated distributions were presented as a density smooth of the simulated data (truncated as the inner 90th percentiles); the diamond centers represent the observed deciles and the left–right extremes provide a relative range of 80–125% of the observed deciles.

Ideally, the simulated density smooth contained the observed value (this served as a visual indicator of minimal bias between observed and simulated values) and was contained within the 80–125% range (this served as a visual indicator of minimal imprecision and lack of marked bias between observed and simulated values). Furthermore, failures to meet these containment criteria were interpreted based on the percentile missed. For example, if the predictive check showed bias near the 10th and/or 90th percentile but was within range for the central percentiles, then the interpretation was that the model was adequately predicting the average, or typical, individual subjects, but was either under- or over-predicting the extreme subjects; this would need to be considered if the model was to be used for additional simulations.

In addition to providing a full comparison of the relative distributions of the observed versus simulated data, this approach affords a fairly simple means for evaluating model performance for relative covariate effects. A weight categorization is included in the current predictive check to illustrate this point. A standard visual predictive check plot[24] was also included for general reference.

## Results

The patient population from the five studies consisted of 974 (574 male, 400 female) patients with a diagnosis of T2DM. Baseline characteristics are summarized in Table 1. Patients contributed 8,289 quantifiable empagliflozin plasma concentrations, comprising 1,116, 2,240, 1,029, 1,390, and 2,514 observations from *Studies A*, *B*, *C*, *D*, and *E*, respectively. The combined study population was mainly white (77%) or Asian (22%; N = 100 and 112 from *Studies C* and *D*, respectively). The mean (min–max) age and weight were 58 (28–80) years and 85 (44–152) kg, respectively; the mean; weight in *Study C* (Japanese patients) was lower (68 [44–98] kg) compared to the other studies. Most (>67%) patients had mean estimated creatinine clearance >90 mL/min; only 14 patients (<1.5%) had estimated creatinine clearance <50 mL/min. Most other laboratory values were within normal ranges (Table 1).

The observed empagliflozin plasma concentrations versus time displayed a bi-exponential decline with a relatively rapid disposition phase followed by a more prolonged disposition phase (Figure 1). Subsequently, the population PK of empagliflozin in patients with T2DM was described by a two-compartmental model with lagged first-order oral absorption (Table 2). Diagnostic plots (Figure 2) enabled goodness-of-fit evaluation of the final full empagliflozin population PK model with the observed data.

The interindividual variance (IIV) estimates were between 15% and 30% CV. Residual variance was estimated to be greater for empagliflozin concentrations collected from *Studies D* and *E* (36% CV) compared to concentrations collected from the other studies (20% CV) (Table 2 and Figure 1). Correlations between the interindividual random effects were relatively small: 0.30, 0.27 and −0.13 between $\eta_{CL/F}$ (ETA1) − $\eta_{V3/F}$ (ETA2), $\eta_{CL/F} - \eta_{KA}$ (ETA3), and $\eta_{V3/F} - \eta_{KA}$, respectively.

The distributions (95th percentiles) of 500 nonparametric bootstrap estimates of the PK model parameter

**Table 1.** Baseline Demographic and Laboratory Information

| Study | A (n = 48) | B (n = 78) | C (n = 100) | D (n = 324) | E (n = 424) |
|---|---|---|---|---|---|
| Male/female, n/n | 39/9 | 67/11 | 84/16 | 172/152 | 212/212 |
| Race, n | | | | | |
| White | 47 | 76 | 0 | 212 | 416 |
| Black | 1 | 1 | 0 | 0 | 6 |
| Asian | 0 | 1 | 100 | 112 | 0 |
| Hawaiian/Pacific | 0 | 0 | 0 | 0 | 2 |
| Age, years | 57 (8.9), (33–68) | 57 (8.7), (34–69) | 57 (9.2), (34–70) | 58 (10.0), (28–80) | 58 (8.6), (32–78) |
| Weight, kg | 95 (14), (68–123) | 93 (15), (69–127) | 68 (13), (44–98) | 81 (17), (46–152) | 89 (16), (55–139) |
| Height, cm | 175 (9.1), (146–198) | 175 (8.2), (158–198) | 166 (8.1), (142–184) | 167 (9.9), (145–196) | 168 (10.1), (138–196) |
| BMI, kg/m$^2$ | 30.8 (3.5), (23.8–29.4) | 30.4 (4.5), (22.8–39.5) | 24.6 (3.8), (17.8–39.1) | 28.9 (4.5), (20.1–39.6) | 31.4 (4.5), (19.6–50.3) |
| BSA, m$^2$ | 2.1 (0.2), (1.6–2.5) | 2.1 (0.2), (1.7–2.6) | 1.8 (0.2), (1.3–2.2) | 1.9 (0.2), (1.3–2.8) | 2.0 (0.2), (1.5–2.6) |
| SCR, mg/dL | 1.1 (0.1), (0.9–1.2) | 0.9 (0.1), (0.5–1.2) | 0.8 (0.1), (0.7–1.0) | 0.9 (0.1), (0.7–1.8) | 0.9 (0.2), (0.6–1.4) |
| $CR_{CL}$, mL/min | 103 (27.2), (64–182) | 117 (30.8), (70–264) | 94 (23.7), (51–159) | 99 (29.3), (39–202) | 107 (31.1), (47–264) |
| ALT, U/L | 37.6 (16.5), (13.4–79.5) | 37.9 (16.2), (12.1–85.5) | 29.4 (20.7), (4.5–151) | 37.1 (26.2), (0–206) | 40.4 (27.8), (4–190) |
| AST, U/L | 23.3 (10.5), (2.6–45.0) | 25.9 (13.4), (5.3–73.8) | 18.1 (16.1), (−2.1–134) | 28.1 (20.9), (2.1–197) | 29.5 (21.4), (−6.4–180) |
| AP, U/L | 124 (54.2), (54.7–303) | 116 (54.8), (36–425) | 127 (41.5), (23.5–250) | 139 (48.8), (28.6–349) | 125 (47.6), (22.1–408) |
| LDH, U/L | 201 (16.6), (176–239) | 194 (54.3), (86.7–575) | 185 (14.4), (156–231) | 195 (19.4), (127–276) | 187 (18.2), (134–259) |
| CK, U/L | 186 (62.2), (119–387) | 174 (57), (49.9–322) | 152 (48.3), (83.4–351) | 226 (129), (78–1180) | 228 (136), (57–1060) |
| GGT, U/L | 43.7 (24.2), (13.0–145) | 44.7 (28.5), (5.4–181) | 40.8 (31.7), (−1.5–156) | 53.2 (66.3), (−3.1–733) | 52.7 (64.5), (−3.1–586) |
| Total protein, g/dL | 6.93 (0.522), (6.0–8.2) | 6.62 (0.981), (−0.536–7.87) | 6.99 (0.378), (6.29–8.31) | 7.42 (0.468), (6.22–9.19) | 7.34 (0.434), (5.62–8.75) |
| Triglycerides, mg/dL | 184 (108), (48.9–542) | 203 (106), (43.1–723) | 112 (53.5), (37.2–314) | 135 (135), (19.6–1740) | 132 (85.1), (15.3–675) |
| Cholesterol, mg/dL | 180 (16.2), (126–225) | 180 (25.1), (106–236) | 126 (29.9), (69.6–205) | 126 (39.6), (27.5–396) | 116 (38.2), (25.3–336) |

BMI, body mass index; BSA, body surface area; SCR, serum creatinine; $CR_{CL}$, estimated creatinine clearance; ALT, alanine transaminase; AST, aspartate transaminase; AP, alkaline phosphatase; LDH, lactate dehydrogenase; CK, creatine kinase; GGT, gamma-glutamyl transferase.
Data are mean (SD) and (range [min–max]) unless specified.

estimates were examined to determine the relative magnitude and estimation precision of each covariate effect (Figure 3). Most covariate effect parameters were estimated with sufficient precision and the CIs were considered to fall within the 75–125% boundaries of the

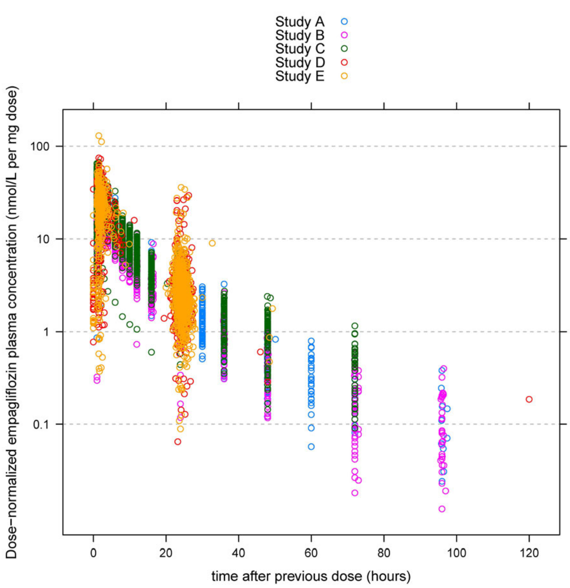


**Figure 1.** Observed overlay of empagliflozin dose-normalized plasma concentrations versus time.

null effect. The major covariate effect was the positive relationship imposed by the allometric scaling between weight and the PK parameters CL/F, V2/F, V3/F, and Q/F (not shown). Relative to the approximate median weight (85 kg), body weight over a 54–123 kg range, which represented 2.5th and 97.5th percentiles of the observed weights, was estimated from the allometric effects to be related to only a ±30% difference in CL/F; this translated to expected steady-state exposures ($AUC_{ss}$) for empagliflozin 25 mg q.d. of 6,830, 4,860, and 3,700 nmol h/L at body weights of 54, 85, and 123 kg, respectively.

The predictive performance of the final population PK model was assessed for model evaluation (Figure 4 and Online Supplementary Figure S1). Evaluation of the relative influence of weight on exposure from the categorization (<85 kg vs. ≥85 kg) of the predictive check revealed that the model appropriately described exposures for both weight categories, and that exposures were typically no more than 25–30% greater in patients weighing <85 kg compared with ≥85 kg.

On average, men in this T2DM patient population weighed more than women, and non-Asian patients weighed more than Asian patients. Therefore, although sex and race themselves did not lend any additional description to PK variability when incorporated into the population PK model as covariates, exposure differences between males and females and between Asian and non-

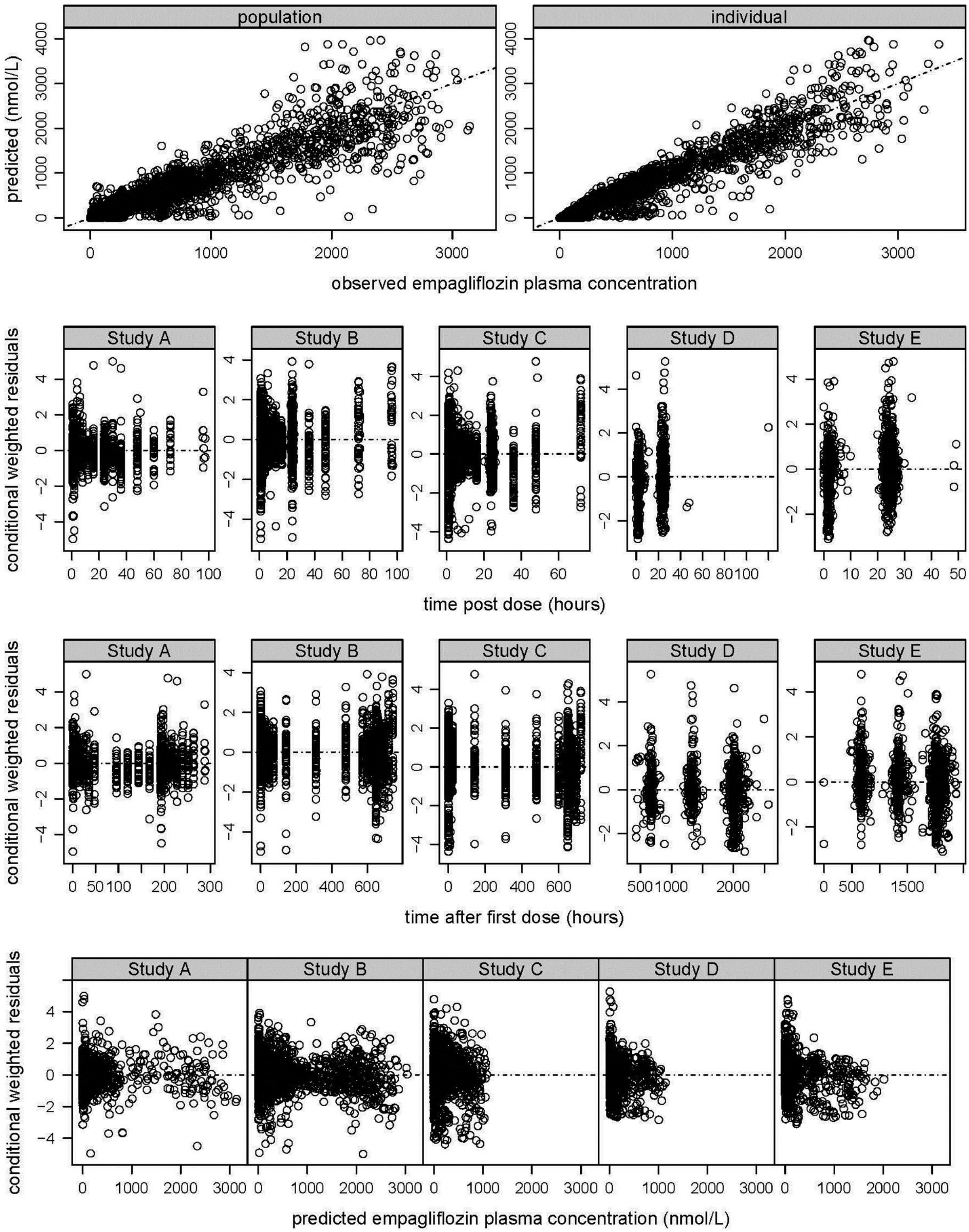


**Figure 2.** Diagnostic plots for the final full empagliflozin population PK model.

Asian patients were considered to be appropriately described by the allometric weight scaling alone.

No additional covariate effects (age, sex, total protein, and smoking and alcohol history) on the PK parameters were considered to affect PK variability (95% CIs for the effect sizes were within 75–125% of a null [zero effect] value). In addition, an extended model was applied which included the effects of liver enzymes (alanine transaminase, alkaline phosphatase, gamma-glutamyl transferase, and lactate dehydrogenase) as covariates on oral

**Table 2.** Population Pharmacokinetic Model Parameter Estimates

| | | Nonparametric bootstrap | | |
|---|---|---|---|---|
| | Point estimate | Median | 95% CI | IIV |
| CL/F ($\theta_1$) | **9.87 (L/h)** | **9.86** | **9.33, 10.4** | **26.9 (CV%)** |
| CL/F ~ (AGE/50)$^{\theta 7}$ | −0.192 | −0.188 | −0.338, −0.0313 | |
| CL/F ~ SMK1($\theta_8$) | 0.991 | 0.991 | 0.953, 1.04 | |
| CL/F ~ SMK2($\theta_9$) | 1.02 | 1.02 | 0.981, 1.07 | |
| CL/F ~ ALC ($\theta_{10}$) | 1.02 | 1.02 | 0.984, 1.06 | |
| CL/F ~ ASIAN ($\theta_{11}$) | 0.98 | 0.98 | 0.938, 1.03 | |
| CL/F ~ (TPRO/6.8)$^{\theta 12}$ | −0.345 | −0.340 | −0.490, −0.190 | |
| CL/F ~ (0.8/SCR)$^{\theta 13}$ | 0.249 | 0.250 | 0.168, 0.334 | |
| CL/F ~ (WT/70)$^{\theta 20}$ | 0.75 | FIXED | | |
| CL/F ~ SEX ($\theta_{24}$) | 0.988 | 0.988 | 0.942, 1.04 | |
| V2/F ($\theta_2$) | **3.02 (L)** | **3.06** | **2.40, 3.76** | |
| V2/F ~ (AGE/50)$\theta^{14}$ | 0.0807 | 0.0838 | −0.0985, 0.256 | |
| V2/F ~ ASIAN ($\theta_{15}$) | 1.20 | 1.21 | 0.925, 1.58 | |
| V2/F ~ (TPRO/6.8)$^{\theta 16}$ | −0.0331 | −0.0406 | −0.292, 0.234 | |
| V2/F ~ (WT/70)$^{\theta 21}$ | 1 | FIXED | | |
| V2/F ~ SEX ($\theta_{25}$) | 0.948 | 0.948 | (0.901, 0.996) | |
| V3/F ($\theta_4$) | **60.4 (L)** | **60.2** | **56.4, 64.3** | **30.8 (CV%)** |
| V3/F ~ (AGE/50)$^{\theta 17}$ | 0.246 | 0.261 | −0.0146, 0.528 | |
| V3/F ~ ASIAN ($\theta_{18}$) | 1.15 | 1.16 | 1.06, 1.28 | |
| V3/F ~ (TPRO/6.8)$^{\theta 19}$ | −0.336 | −0.336 | −0.678, −0.0217 | |
| V3/F ~ (WT/70)$^{\theta 23}$ | 1 | FIXED | | |
| V3/F ~ SEX ($\theta_{26}$) | 1.02 | 1.02 | 0.916, 1.12 | |
| Q/F ($\theta_3$) | **5.16 (L/h)** | **5.16** | **4.82, 5.56** | |
| Q/F ~ (WT/70)$^{\theta 22}$ | 0.75 | FIXED | | |
| KA ($\theta_5$) | **0.224 ($h^{-1}$)** | **0.227** | **0.176, 0.279** | **15.2 (CV%)** |
| KA ~ ASIAN ($\theta_{27}$) | 1.24 | 1.23 | (0.939, 1.65) | |
| Residual variance | | | | |
| Studies A, B, and C | | | | |
| CV% | 19.6 | 19.6 | 18.5, 20.8 | |
| SD (nmol/L) | 0.179 | 0.19 | 0.010, 0.334 | |
| Studies D and E | | | | |
| CV% | 35.7 | 35.9 | 33.7, 38.0 | |
| SD (nmol/L) | 0.010 | 0.010 | 0.010, 0.010 | |

IIV, interindividual variance.
An oral absorption lag time (ALAG1) was included and fixed at 0.5 hours.
Bold values indicate the reference population PK parameter estimates.

clearance. Point estimates for these additional effects were all 1.00 (unity).

## Discussion

The goals of this analysis were to describe empagliflozin PK and to estimate the effects of covariates on the variability of empagliflozin PK. An imposed allometric weight effect was determined to be the most influential PK covariate effect. Covariate–parameter relationships were investigated using a prespecified full model approach,[25–29] rather than a stepwise model approach. In a stepwise approach, forward or backward comparisons are made across multiple models, based on the likelihood ratio test[30] and a prespecified $\alpha$-level, with each model expressing different covariate–parameter combinations.[31,32] The prespecified full model approach emphasizes parameter estimation. Using this approach, predefined covariate–parameter relationships were identified based on exploratory graphics, scientific interest, and mechanistic plausibility.

Covariate–parameter relationships were assessed visually through covariate plots generated to include the typical PK parameter (e.g., CL/F point estimate and 95% CI), and the expected value (and 95% CI) for each categorical covariate and for select values of continuous predictors (e.g., near the lower, midpoint, and upper ranges of weight). As an example, Figure 3A depicts several scenarios of covariate effects on CL/F: the expected changes associated with weight; a true lack of effect between (1) male and female, and (2) Asian and non-Asian patients; and several effects (e.g., age, total protein, serum creatinine) that deviate from the values for the typical individual but are wholly contained within 75–

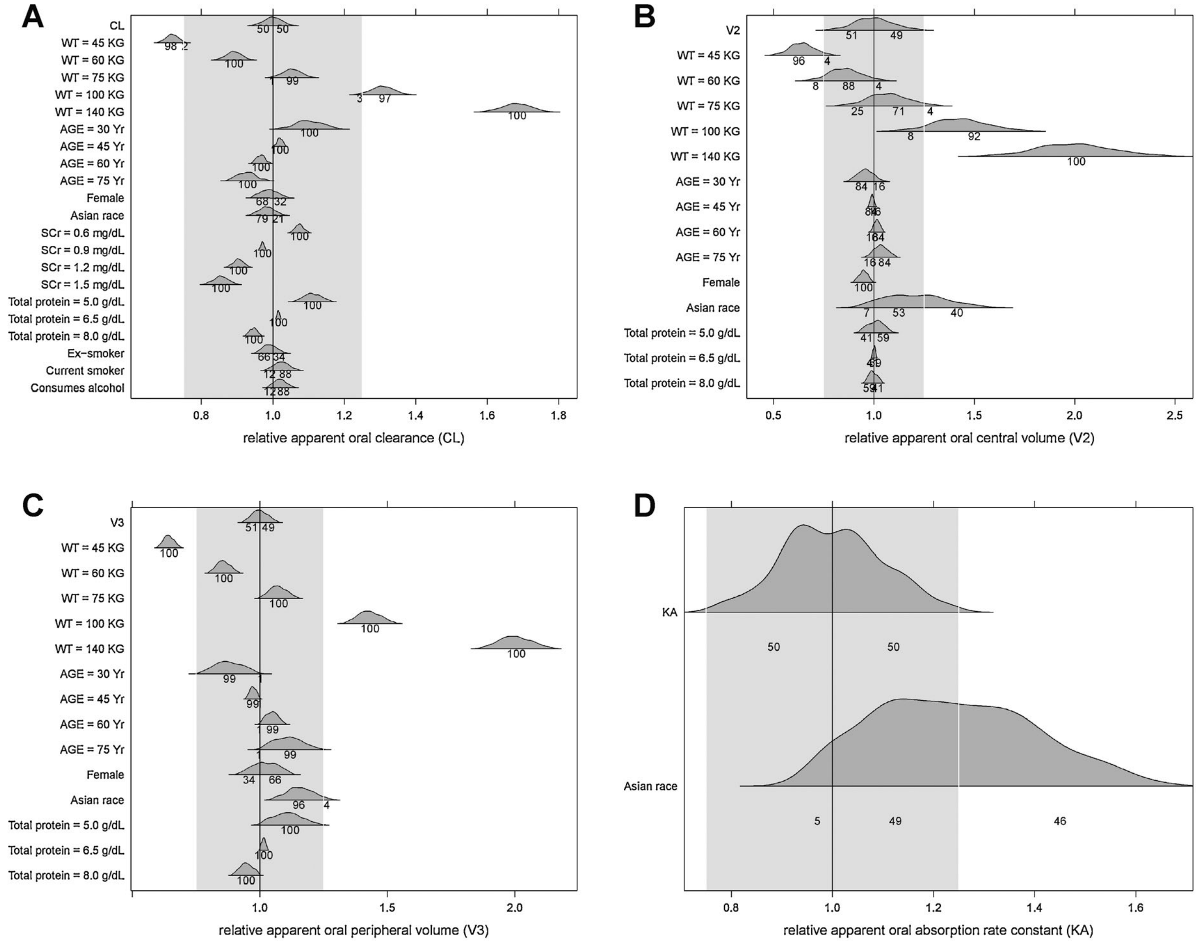


**Figure 3.** Distributions of 500 nonparametric bootstrap estimates for the final full empagliflozin population PK model. The hypothetical reference individual is defined as a 50 year old, non-smoking Caucasian man with a body weight of 70 kg, no history of alcohol use, total protein of 6.8 mg/dL, and serum creatinine of 0.8 g/dL. The horizontal lines represent the 95th percentiles of the bootstrap estimates. The distributions (95th percentiles) of the bootstrap estimates are provided as density smooths atop each horizontal range. All evaluations were normalized by dividing by the named PK parameter's point estimate, such that a value of 1 represents a null covariate effect. The gray-shaded areas represent the null value $\pm$25% and are intended to reflect areas in which the covariate effect represents a minimal change from the null effect (i.e., not expected to be clinically meaningful). The numbers on the plot for each effect estimate represent, from left to right, the percentage of the 500 nonparametric bootstrap estimates that were $<-25\%$, between $-25\%$ and the null, between the null and 25%, and $>25\%$ of the null. For continuous covariates, estimates are provided at lower, intermediary, and upper values of the covariate's range. Results are shown for (A) apparent clearance (CL), (B) apparent central volume of distribution (V2), (C) apparent peripheral volume of distribution (V3), and (D) absorption rate constant (KA).

125% of these values. Similar relationships hold for the distributive volume covariates. An exception is the lack of information about the effect of Asian race on central volume (V2/F, Figure 3B) where the lack of a precise estimate is indicated by the distribution of estimates extending beyond the "null" range. By retaining this as a covariate, however, acknowledgement and quantification of the lack of information were provided, which contrasts with a statement of "insignificance" that would have occurred during stepwise model development, due to a lack of power to detect a difference for such a small group. The covariate analysis also indicated that Asian patients might experience a slightly increased rate of oral empagliflozin absorption compared to non-Asian patients with T2DM. This was due to the approximately 25% greater oral absorption rate constant for Asian patients. Rate of absorption is associated with $C_{max}$ and the time to $C_{max}$ ($T_{max}$) and so should be considered in any subsequent applications that consider any associations with empagliflozin $C_{max}$ or $T_{max}$.

The values used as references for the categorical covariates and to normalize the continuous covariates corresponded to a hypothetical reference individual, defined as a 50 year old, non-smoking Caucasian man with a body weight of 70 kg, no history of alcohol use, total protein of 6.8 mg/dL, and serum creatinine of 0.8 g/

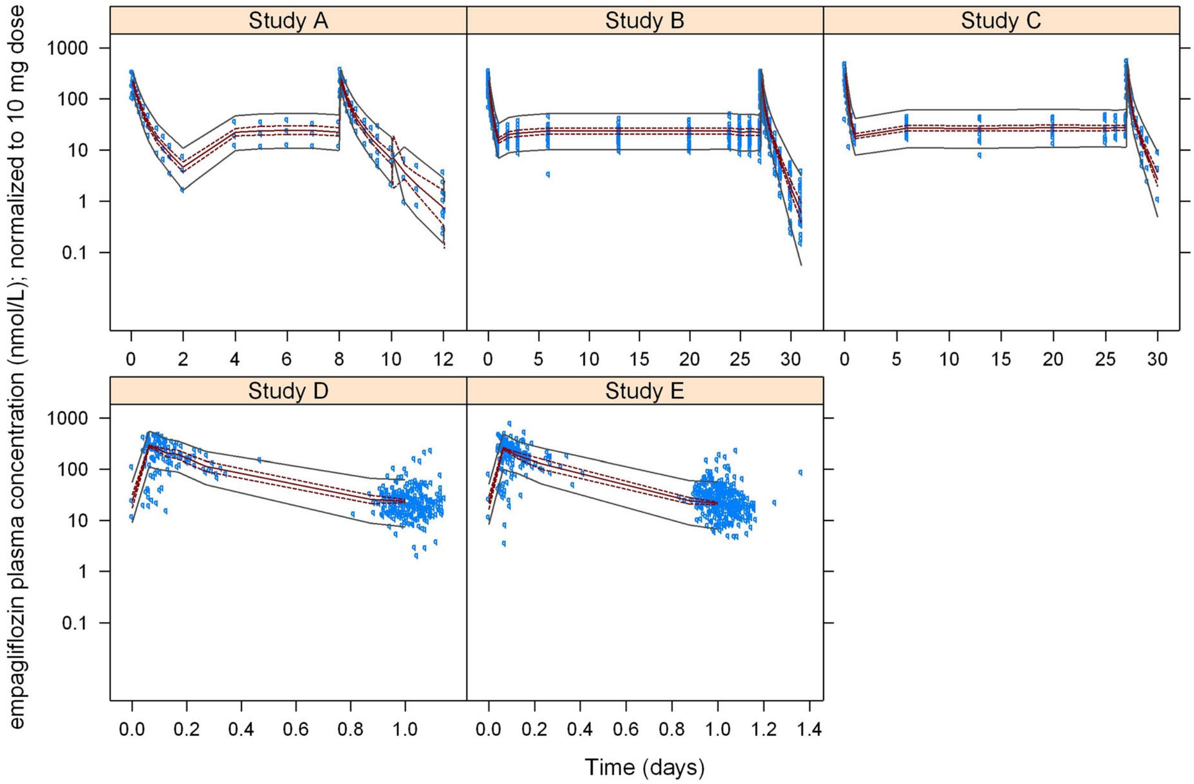


**Figure 4.** Visual predictive check of the final full empagliflozin population PK model. Observed (blue circles) plasma concentration-time profiles (normalized to a 10 mg dose) with an overlay of the median (red solid line) and the 5th and 95th percentiles (solid gray lines) of the Monte Carlo-simulated values. The 90% confidence band for the median is also shown (red dashed lines). Time for *Studies A, B,* and *C* is the time after the first dose; time for *Studies D* and *E* is time relative to the previous dose.

dL; these characteristics correspond to an individual with an estimated creatinine clearance of 116 mL/min. The PK parameter point estimates (Table 2) equated to disposition half-lives of 0.1 ($t_{1/2,\alpha}$) and 12 ($t_{1/2,\beta}$) hours, respectively, for the reference individual. The reference absorption half-life estimate, which was constrained to be greater than the typical population $t_{1/2,\alpha}$, was 3.1 hours.

Goodness-of-fit criteria (Figure 2) revealed that the final model was consistent with the observed data and that no systematic bias remained. The model evaluation results provided evidence that both the fixed- and random-effects components of the final model were reflective of the observed data.

Low interindividual variance (IIV) estimates (15%–30% CV), indicated minimal between-subject differences in the extent of empagliflozin plasma exposure for a given oral dose. These variability estimates, along with the lack of marked covariate effects suggests that no exposure-based dose adjustments were required within the study population and investigated dose range.

Minimal correlations between interindividual random effects indicate that there were no issues with unique identification of random effects. However, η-shrinkage for $\eta_{CL/F}$ (25.6%), $\eta_{V3/F}$ (57.9%) and $\eta_{KA}$ (47.4%) may have masked true correlations.[33] The lower η-shrinkage for $\eta_{CL/F}$ (25.6%) is relevant because CL/F is the determinant for AUC, supporting the use of Bayesian MAP individual AUC estimates used for subsequent exposure–response analyses.

Model-derived simulations were in agreement with the observed data, as judged by visual inspection of the predictive check plots (Figure 4), and the model was shown to appropriately describe exposures for the weight categories <85 and ≥85 kg (Figure S1). Age, sex, total protein, and smoking and alcohol history were not considered to affect PK variability. The extended model indicated that liver enzymes did not markedly affect empagliflozin exposure, although an additional evaluation with a broader range of data may be required to define the precision of these estimates.

Overall, the final population PK model presented herein, which included the pre-specified covariate effects, appropriately described the empagliflozin plasma concentrations from the five clinical studies investigated, as evidenced by the general goodness-of-fit diagnostics and by the predictive check model evaluation. The model provided an improved understanding of the PK properties of empagliflozin, its associated PK variability, and covariate predictors of this variability following multiple oral dosing to patients with T2DM and may be used to predict exposures for subsequent evaluations of safety and efficacy exposure–response.

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## Supporting Information

Additional supporting information may be found in the online version of this article at the publisher's web-site.

**Figure S1.** Predictive check of the final full empagliflozin population PK model categorized by weight. Deciles of individual-specific minimum ($C_{min}$), median ($C_{median}$) and maximum ($C_{max}$) observed empagliflozin plasma concentration are compared for observed data (blue diamonds) and for 300 Monte Carlo simulations from the final model. Left-right extremes of diamonds are 0.8 and 1.25 times the diamond centers (observed decile). Polygons are density smooths of the simulated deciles.